EXPERIMENT 5: EFFECT OF ROUTE OF ADMINISTRATION ON DRUG EFFECT

300 Level Pharmacology Practical & In-Course Examination Master Study Guide

Department of Pharmacology, College of Medicine / University Teaching Hospital
Source Document: Practical Lab Manual & Report — Page 41 (Printed Experimental Text)
Target: 30 Verbatim High-Yield Multiple Choice Questions (MCQs) & In-Course Rationales

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📄 Exact Verbatim Transcription of Page 41 (Printed Text)

For complete reference and verbatim verification, here is the exact text printed on Page 41 of the laboratory manual:

EXPERIMENT 5

EFFECT OF ROUTE OF ADMINISTRATION ON DRUG EFFECT

Aim: To find out the effect that is produced when the same drug is administered into the body via different routes.
Animals: Rats -Male
Materials: Syringes and needles, Oral feeding tube, stop watch.
Drugs: Pentobarbitone or Nembutal
Method: Weigh the first animal and inject 100mg/kg Pentobarbital intraperitoneally. Start the stopwatch and observe the animal, turn it to lie on its back until the animal is not able to right itself. Record the time of loss of righting reflex.

Repeat the above procedure after:
a. Oral feeding of 100mg/kg
b. i/p injection of 100mg/kg
c. i/m injection of 100mg/kg
d. s/c injection of 100mg/kg

Answer the following questions:
1. What is the time of onset of effect by the various routes?
2. What is the duration of the drug action?
3. What factors may be responsible for the differences observed in the time of onset of effect of drugs by the various routes?

🎯 High-Yield Conceptual Summary for In-Course Exams

1. **Why Route Matters (Pharmacokinetic Determinants of Onset & Duration):**
   - **Intraperitoneal (IP):** Very rapid onset due to massive surface area and rich vascular/mesenteric capillary networks of the peritoneal cavity. *(Note: Drugs absorbed IP enter portal circulation and*

undergo first-pass hepatic metabolism).

- **Intramuscular (IM):** Rapid onset; rate depends on vascular perfusion of the specific muscle group (deltoid > vastus lateralis > gluteus maximus).
- **Subcutaneous (SC):** Slower, sustained onset due to lower vascularity and higher adipose tissue resistance compared to muscle.
- **Oral (PO):** Slowest onset; delayed by gastric emptying time, dilution by gastric contents, intestinal mucosal barrier transit, and obligatory hepatic first-pass metabolism.

2. **Standardized Clinical Endpoint: Loss of Righting Reflex (LORR):**
   - The **righting reflex** is an involuntary motor reflex mediated by the vestibular apparatus, cerebellum, and brainstem reticular formation that restores a quadruped to its feet when placed on its dorsum.
   - **Loss of Righting Reflex (LORR)** is the gold-standard preclinical surrogate marker for the onset of surgical anesthesia and loss of consciousness in laboratory rodents.
3. **Pentobarbitone (Nembutal) Pharmacology:**
   - Short-acting oxybarbiturate.
   - **Mechanism of Action:** Binds to allosteric sites on the neuronal \text{GABA}_\text{A} receptor complex → prolongs the open duration of chloride (Cl^-) ion channels → causes membrane hyperpolarization → depresses CNS activity. At high concentrations, it directly opens chloride channels and depresses excitatory AMPA receptors.
4. **Methodological Controls:**
   - **Use of Male Rats:** Eliminates cyclical variations in hepatic cytochrome P450 enzyme activity and plasma binding proteins driven by the 4–5 day rodent estrous cycle (estrogen and progesterone).
   - **Weight Standardization:** Dose is strictly fixed at **100 mg/kg** across all routes to isolate the *route of administration* as the sole independent variable.

📝 The 30 Verbatim In-Course Multiple Choice Questions (MCQs)

Question 1: Official Experiment Designation

What is the exact numerical designation and title of this laboratory experiment as printed at the top of Page 41?

- A) Experiment 3: Individual Variations in the Response to Drugs
- B) Experiment 4: Bioassay of Histamine on Isolated Guinea Pig Ileum
- C) Experiment 5: Effect of Route of Administration on Drug Effect
- D) Experiment 6: Potentiation of Barbiturate Sleeping Time by Chlorpromazine

Correct Answer: `C`

- **Verbatim Text on Page 41:** `EXPERIMENT 5 / EFFECT OF ROUTE OF ADMINISTRATION ON DRUG EFFECT`

- **In-Course Explanation:** Practical exams require accurate identification of the experiment number and official title. Experiment 5 investigates how varying routes alter pharmacokinetics.
- **Distractor Breakdown:** Option A is Experiment 3 (Individual Variations); Option B is an in vitro organ bath bioassay; Option D is an enzyme inhibition/synergism practical.

Question 2: Stated Experimental Aim

According to the printed text on Page 41, what is the stated Aim of Experiment 5?

- A) To determine the median lethal dose (LD_{50}) of pentobarbital in healthy male rats
- B) To find out the effect that is produced when the same drug is administered into the body via different routes
- C) To compare the potency of pentobarbitone against diazepam in producing skeletal muscle relaxation
- D) To measure the rate of hepatic microsomal enzyme induction following oral feeding

Correct Answer: `B`

- **Verbatim Text on Page 41:** `Aim: To find out the effect that is produced when the same drug is administered into the body via different routes.`
- **In-Course Explanation:** The experiment uses a single drug (pentobarbitone) at a constant weight-adjusted dose (100 mg/kg) to compare onset, peak effect, and duration across routes.
- **Distractor Breakdown:** Option A is a quantal lethality bioassay; Option C evaluates comparative potency of two different chemical entities; Option D measures enzyme induction kinetics.

Question 3: Animal Species Specified

What specific laboratory animal species is mandated under the "Animals" section of Page 41?

- A) Guinea pigs
- B) Albino mice
- C) Rats
- D) New Zealand white rabbits

Correct Answer: `C`

- **Verbatim Text on Page 41:** `Animals: Rats -Male`
- **In-Course Explanation:** Rats (*Rattus norvegicus*) are specified for this experiment because their size allows straightforward righting reflex testing and accurate intramuscular/intraperitoneal dosing.
- **Distractor Breakdown:** Option A (guinea pigs) is used for isolated ileum and tracheal chain bioassays; Option B (mice) was used in Experiment 3; Option D (rabbits) is used for pyrogen testing and mydriasis.

Question 4: Gender Selection of Test Animals

What specific sex of rats is designated on Page 41?

- A) Female
- B) Male

- C) Equal proportions of male and female
- D) Pregnant female

Correct Answer: `B`

- **Verbatim Text on Page 41:** `Animals: Rats -Male`
- **In-Course Explanation:** Male rats are universally preferred in baseline pharmacokinetic experiments to avoid hormonal variations associated with the estrous cycle.
- **Distractor Breakdown:** Option A and C introduce hormonal confounding; Option D is used in teratogenicity assays.

Question 5: Pharmacological Rationale for Animal Gender Selection

In experimental pharmacology, what is the primary scientific rationale for using male rats rather than female rats when comparing routes of administration?

- A) Male rats completely lack cytochrome P450 enzymes
- B) Female hormonal fluctuations (estrous cycle) cause variable hepatic drug metabolism and volume of distribution
- C) Male rats do not possess a righting reflex
- D) Female rats cannot absorb drugs through the peritoneal cavity

Correct Answer: `B`

- **In-Course Explanation:** The 4-to-5-day rodent estrous cycle causes cyclical surges of estrogen and progesterone, which modulate hepatic CYP450 enzyme expression and plasma protein levels. Using exclusively male rats minimizes intra-group pharmacokinetic variability.
- **Distractor Breakdown:** Option A is false (males have abundant CYP450s); Option C is false (all quadrupeds possess righting reflexes); Option D is biologically impossible.

Question 6: Apparatus for Injections

Which apparatus is specified on Page 41 for administering parenteral injections to the animals?

- A) Syringes and needles
- B) Automated micro-infusion pump
- C) Trocar and cannula
- D) Micropipette with plastic tips

Correct Answer: `A`

- **Verbatim Text on Page 41:** `Materials: Syringes and needles, Oral feeding tube, stop watch.`
- **In-Course Explanation:** Calibrated syringes fitted with hypodermic needles are required for the three parenteral routes (IP, IM, and SC).
- **Distractor Breakdown:** Option B is for continuous IV infusions; Option C is for laparoscopic trocars; Option D is for in vitro fluid dispensing.

Question 7: Specialized Apparatus for Enteral Administration

Which specific item in the Materials list is used to administer the drug orally to the rats?

- A) Intravenous catheter
- B) Oral feeding tube
- C) Nasogastric Levin tube
- D) Gastrostomy button

Correct Answer: `B`

- **Verbatim Text on Page 41:** `Materials: Syringes and needles, Oral feeding tube, stop watch.`
- **In-Course Explanation:** An oral feeding tube (gavage needle with a smooth, blunted bulbous tip) is passed down the esophagus directly into the stomach, preventing aspiration into the trachea.
- **Distractor Breakdown:** Option A is for intravascular access; Option C is human-specific clinical equipment; Option D is a surgical enteral feeding port.

Question 8: Timing Instrument

What specific timing device is explicitly mandated in the Materials list on Page 41?

- A) Sand timer
- B) Pendulum clock
- C) stop watch
- D) Electronic kymograph timer

Correct Answer: `C`

- **Verbatim Text on Page 41:** `Materials: Syringes and needles, Oral feeding tube, stop watch.`
- **In-Course Explanation:** A stopwatch is started the instant the injection is completed to record exact latency until the righting reflex is lost.
- **Distractor Breakdown:** Options A and B lack precision; Option D records mechanical drum rotation speed in organ bath experiments.

Question 9: Listed Generic Drug

What is the primary generic drug name listed under the "Drugs" heading on Page 41?

- A) Diazepam
- B) Pentobarbitone
- C) Phenobarbitone
- D) Thiopentone

Correct Answer: `B`

- **Verbatim Text on Page 41:** `Drugs: Pentobarbitone or Nembutal`
- **In-Course Explanation:** Pentobarbitone (British Approved Name) is the designated agent on Page 41.

- **Distractor Breakdown:** Option A is a benzodiazepine; Option C is a long-acting barbiturate anticonvulsant; Option D is an ultra-short-acting IV induction thiobarbiturate.

Question 10: Proprietary / Brand Name

What alternative trade/brand name is provided alongside Pentobarbitone under "Drugs" on Page 41?

- A) Valium
- B) Luminal
- C) Nembutal
- D) Seconal

Correct Answer: `C`

- **Verbatim Text on Page 41:** `Drugs: Pentobarbitone or Nembutal`
- **In-Course Explanation:** Nembutal is the classic brand name for pentobarbital sodium (coined from Na + **e**thyl + **m**ethyl + **but**yl + **al**).
- **Distractor Breakdown:** Option A is diazepam; Option B is phenobarbital; Option D is secobarbital.

Question 11: Duration of Action Classification of Pentobarbitone

To which pharmacological class based on duration of action does Pentobarbitone (Nembutal) belong?

- A) Ultra-short acting barbiturate (duration < 30 minutes)
- B) Short-to-intermediate acting barbiturate (duration 2 to 4 hours)
- C) Long-acting barbiturate (duration 1 to 2 days)
- D) Non-sedating barbiturate analog

Correct Answer: `B`

- **In-Course Explanation:** Pentobarbitone and amobarbitone are classified as short-to-intermediate acting barbiturates (elimination half-life 15–48 hours, anesthetic action 2–4 hours). Thiopentone is ultra-short acting; phenobarbitone is long-acting.
- **Distractor Breakdown:** Option A describes thiopentone and methohexital; Option C describes phenobarbitone; Option D is pharmacologically incorrect.

Question 12: Molecular Mechanism of Action of Barbiturates

At the molecular level, how does Pentobarbitone depress central nervous system activity?

- A) Blocks voltage-gated sodium channels in the peripheral nodes of Ranvier
- B) Binds to \text{GABA}_\text{A} receptors and increases the duration of chloride (Cl^-) channel opening
- C) Competitively blocks post-synaptic muscarinic M_1 receptors in the cerebral cortex
- D) Inhibits monoamine oxidase A (MAO-A) to deplete norepinephrine

Correct Answer: `B`

- **In-Course Explanation:** *Classic exam point:* Barbiturates increase the **duration** of \text{GABA}_\text{A} chloride channel opening ("Barbi-DURATE-ion"), whereas benzodiazepines increase channel opening **frequency** ("Fren-ZO-diazepine").
- **Distractor Breakdown:** Option A is the mechanism of local anesthetics; Option C describes atropine/anticholinergics; Option D describes MAO inhibitors.

Question 13: Initial Step in the Experimental Method

What is the very first procedural action instructed in the Method on Page 41 before administering any drug?

- A) Place the animal in warm water
- B) Weigh the first animal
- C) Shave the rat's abdomen
- D) Fast the animal for 48 hours

Correct Answer: `B`

- **Verbatim Text on Page 41:** `Method: Weigh the first animal and inject 100mg/kg Pentobarbital intraperitoneally.`
- **In-Course Explanation:** The animal must be weighed first so the exact volume of drug solution matching the target dose (100 mg/kg) can be calculated.
- **Distractor Breakdown:** Options A, C, and D are not listed in the protocol.

Question 14: Standard Experimental Dose

What exact standardized dose of Pentobarbital is specified for administration across all routes?

- A) 10 mg/kg
- B) 25 mg/kg
- C) 50 mg/kg
- D) 100mg/kg

Correct Answer: `D`

- **Verbatim Text on Page 41:** `inject 100mg/kg Pentobarbital intraperitoneally`
- **In-Course Explanation:** 100 mg/kg is the standard hypnotic/anesthetic dose of pentobarbital used to produce loss of righting reflex in rats.
- **Distractor Breakdown:** Option A was the amobarbital dose in Experiment 3; Option B was the starting dose in Experiment 3; Option C was an intermediate escalation dose.

Question 15: Primary Injection Route in Method

By which anatomical route is the 100 mg/kg Pentobarbital first injected in the Method?

- A) Intravenously (IV)
- B) Intraperitoneally

- C) Intracerebroventricularly (ICV)
- D) Sublingually

Correct Answer: `B`

- **Verbatim Text on Page 41:** `inject 100mg/kg Pentobarbital intraperitoneally.`
- **In-Course Explanation:** Intraperitoneal (IP) administration delivers drug into the lower abdominal quadrant, ensuring rapid uptake across the peritoneal membrane.
- **Distractor Breakdown:** Option A is difficult in conscious unrestrained rats without a tail-vein cannulation setup; Options C and D are not specified.

Question 16: Immediate Action Following Drug Injection

What must the investigator do immediately after completing the injection, according to Page 41?

- A) Feed the animal glucose water
- B) Start the stopwatch and observe the animal
- C) Return the animal to a dark storage room
- D) Draw a retro-orbital blood sample

Correct Answer: `B`

- **Verbatim Text on Page 41:** `Start the stopwatch and observe the animal, turn it to lie on its back until the animal is not able to right itself.`
- **In-Course Explanation:** Timing must commence at the exact moment of injection to capture the onset latency accurately.
- **Distractor Breakdown:** Options A, C, and D disrupt the observation protocol.

Question 17: Postural Maneuver Performed During Observation

What specific physical maneuver does the investigator perform on the rat during observation?

- A) Suspend the animal by its tail
- B) Place the animal on a heated wire grid
- C) turn it to lie on its back
- D) Pinch the animal's ear with forceps

Correct Answer: `C`

- **Verbatim Text on Page 41:** `turn it to lie on its back until the animal is not able to right itself.`
- **In-Course Explanation:** Turning the animal onto its back tests the vestibular-spinal postural reflex. Normal animals right themselves immediately; anesthetized animals remain supine.
- **Distractor Breakdown:** Option A tests traction or tail-hang reflex; Option B tests hot-plate analgesia; Option D tests nociceptive response.

Question 18: Criterion for Cessation of the Righting Maneuver

Until what specific behavioral endpoint is the animal turned onto its back?

- A) Until the animal's heart rate drops below 60 bpm
- B) until the animal is not able to right itself
- C) Until the animal begins to vocalize
- D) Until pupillary constriction occurs

Correct Answer: `B`

- **Verbatim Text on Page 41:** `until the animal is not able to right itself.`
- **In-Course Explanation:** The test is repeated until the animal completely loses the motor ability to turn from a supine to a prone/quadrupedal stance.
- **Distractor Breakdown:** Options A, C, and D are not the defined behavioral criteria.

Question 19: Official Experimental Endpoint Recorded

What exact measurement is documented at the conclusion of Step 1 on Page 41?

- A) Record the volume of urine excreted
- B) Record the time of loss of righting reflex.
- C) Record the decrease in body temperature
- D) Record the maximal blood pressure reduction

Correct Answer: `B`

- **Verbatim Text on Page 41:** `Record the time of loss of righting reflex.`
- **In-Course Explanation:** The time from injection to the loss of righting reflex represents the **onset of action** for that specific route.
- **Distractor Breakdown:** Options A, C, and D are unrelated to righting reflex assessment.

Question 20: Preclinical Significance of Loss of Righting Reflex (LORR)

In neuropharmacology and anesthetic screening, what does "Loss of Righting Reflex (LORR)" evaluate?

- A) Death of the animal
- B) Loss of consciousness / onset of surgical hypnotic anesthesia
- C) Development of localized limb paralysis from nerve damage
- D) Complete spinal cord transection

Correct Answer: `B`

- **In-Course Explanation:** Loss of righting reflex is the universal surrogate for general anesthesia in animal models, reflecting bilateral depression of the ascending reticular activating system (ARAS).
- **Distractor Breakdown:** Option A is mortality (LD_{50}); Option C is localized peripheral neurotoxicity; Option D is mechanical trauma.

Question 21: First Comparative Route in Repeating Procedure

Under "Repeat the above procedure after", what is the route and dose listed under option 'a'?

- A) a. Intrathecal injection of 50mg/kg
- B) a. Oral feeding of 100mg/kg
- C) a. Rectal suppository of 100mg/kg
- D) a. Transdermal patch of 25mg/kg

Correct Answer: `B`

- **Verbatim Text on Page 41:** `a. Oral feeding of 100mg/kg`
- **In-Course Explanation:** The oral route is evaluated using gavage delivery of 100 mg/kg pentobarbital into the stomach.
- **Distractor Breakdown:** Options A, C, and D are not listed on Page 41.

Question 22: Second Comparative Route

What route and dose are designated under option 'b' of the repeating procedure on Page 41?

- A) b. i/v injection of 100mg/kg
- B) b. i/p injection of 100mg/kg
- C) b. Inhalation of 2% vapor
- D) b. Epidural infusion of 10mg/kg

Correct Answer: `B`

- **Verbatim Text on Page 41:** `b. i/p injection of 100mg/kg`
- **In-Course Explanation:** `i/p` designates intraperitoneal injection.
- **Distractor Breakdown:** Option A (intravenous) is omitted from this rat protocol due to technical difficulty without cannulation; Options C and D are not listed.

Question 23: Third Comparative Route

What route and dose are listed under option 'c' on Page 41?

- A) c. i/m injection of 100mg/kg
- B) c. Intra-arterial injection of 100mg/kg
- C) c. Sublingual drop of 100mg/kg
- D) c. Topical skin application of 100mg/kg

Correct Answer: `A`

- **Verbatim Text on Page 41:** `c. i/m injection of 100mg/kg`
- **In-Course Explanation:** `i/m` designates intramuscular injection, typically into the quadriceps femoris of the hind leg.
- **Distractor Breakdown:** Options B, C, and D are not evaluated in this experiment.

Question 24: Fourth Comparative Route

What route and dose are listed under option 'd' on Page 41?

- A) d. s/c injection of 100mg/kg
- B) d. Intra-articular injection of 100mg/kg
- C) d. Corneal instillation of 100mg/kg
- D) d. Intracardiac puncture of 100mg/kg

Correct Answer: `A`

- **Verbatim Text on Page 41:** `d. s/c injection of 100mg/kg`
- **In-Course Explanation:** `s/c` designates subcutaneous injection into the loose skin fold over the scruff of the neck.
- **Distractor Breakdown:** Options B, C, and D are not listed in this protocol.

Question 25: The Four Tested Routes Summary

Which complete list of four administration routes is formally compared in Experiment 5?

- A) Intravenous, Intrathecal, Sublingual, and Transdermal
- B) Oral feeding, Intraperitoneal (i/p), Intramuscular (i/m), and Subcutaneous (s/c)
- C) Inhalation, Rectal, Intra-arterial, and Epidural
- D) Nasal spray, Otic drops, Conjunctival, and Buccal

Correct Answer: `B`

- **Verbatim Text on Page 41:** `a. Oral feeding of 100mg/kg / b. i/p injection of 100mg/kg / c. i/m injection of 100mg/kg / d. s/c injection of 100mg/kg`
- **In-Course Explanation:** The experiment systematically contrasts one enteral route (oral) against three parenteral routes (IP, IM, SC).
- **Distractor Breakdown:** Options A, C, and D list alternative clinical routes not tested in this laboratory manual protocol.

Question 26: First Post-Lab Question Printed on Page 41

What is the exact text of Question 1 printed under "Answer the following questions"?

- A) What is the chemical structure of pentobarbitone?
- B) What is the time of onset of effect by the various routes?
- C) What is the LD50 of pentobarbital in male rats?
- D) Which liver enzyme metabolizes barbiturates?

Correct Answer: `B`

- **Verbatim Text on Page 41:** `1. What is the time of onset of effect by the various routes?`

- **In-Course Explanation:** Question 1 focuses on onset time (latency to LORR), which directly correlates with the speed of absorption across different biological membranes.
- **Distractor Breakdown:** Options A, C, and D are not the printed questions on Page 41.

Question 27: Second Post-Lab Question Printed on Page 41

What is the exact text of Question 2 printed under "Answer the following questions"?

- A) What is the molecular weight of Nembutal?
- B) What is the cost per gram of the drug?
- C) What is the duration of the drug action?
- D) What is the volume of distribution in liters?

Correct Answer: `C`

- **Verbatim Text on Page 41:** `2. What is the duration of the drug action?`
- **In-Course Explanation:** Question 2 addresses the duration of action (time from LORR until recovery of the righting reflex), reflecting drug redistribution and elimination.
- **Distractor Breakdown:** Options A, B, and D are not printed on Page 41.

Question 28: Third Post-Lab Question Printed on Page 41

What is the exact text of Question 3 printed under "Answer the following questions"?

- A) What factors may be responsible for the differences observed in the time of onset of effect of drugs by the various routes?
- B) How many animals died during the practical session?
- C) Why is pentobarbitone classified as a Schedule II substance?
- D) What are the signs of acute barbiturate poisoning in humans?

Correct Answer: `A`

- **Verbatim Text on Page 41:** `3. What factors may be responsible for the differences observed in the time of onset of effect of drugs by the various routes?`
- **In-Course Explanation:** Question 3 requires students to explain the physiological and physicochemical determinants of drug absorption (vascularity, surface area, barriers, first-pass metabolism).
- **Distractor Breakdown:** Options B, C, and D are extraneous clinical questions not printed on Page 41.

Question 29: Pharmacological Determinants of Onset of Action

In answering Question 3 of the lab report, which physiological factor is primarily responsible for the fastest onset of action observed with intraperitoneal (i/p) and intramuscular (i/m) routes compared to subcutaneous and oral routes?

- A) Absence of red blood cells in muscle tissue
- B) Rich capillary blood supply and large absorptive surface area at the injection site

- C) Immediate renal excretion of the drug before reaching the brain
- D) Permanent covalent binding of pentobarbitone to skin keratinocytes

Correct Answer: `B`

- **In-Course Explanation:** *High-yield exam rationale:* The rate of absorption is directly proportional to vascular perfusion and surface area. Peritoneal and muscle capillary beds have high perfusion, resulting in rapid systemic uptake, whereas subcutaneous tissue has lower blood flow, and the oral route faces gastrointestinal transit barriers and first-pass hepatic metabolism.
- **Distractor Breakdown:** Option A is false (muscle is highly vascularized); Option C would prevent drug action; Option D is biologically impossible.

Question 30: Expected Clinical Ranking of Onset Speed Across the Four Routes

Based on established pharmacological principles tested in Question 1, what is the expected order of onset of drug action (from fastest onset to slowest onset) for these four routes?

- A) Oral > Subcutaneous > Intramuscular > Intraperitoneal
- B) Intraperitoneal (i/p) > Intramuscular (i/m) > Subcutaneous (s/c) > Oral
- C) Subcutaneous > Oral > Intraperitoneal > Intramuscular
- D) Oral > Intraperitoneal > Subcutaneous > Intramuscular

Correct Answer: `B`

- **In-Course Explanation:** *Core in-course rule:*
  $\text{Intravenous (IV)} > \text{Intraperitoneal (IP)} > \text{Intramuscular (IM)} > \text{Subcutaneous (SC)} > \text{Oral (PO)}$
  IP is the fastest among the four due to the massive peritoneal surface area, followed closely by vascular muscle tissue (IM). Subcutaneous tissue is slower due to poorer perfusion, while oral is the slowest due to gastric emptying delays and gastrointestinal mucosal absorption.
- **Distractor Breakdown:** Options A, C, and D place oral or subcutaneous before intraperitoneal/intramuscular, which contradicts fundamental pharmacokinetic principles.

🏆 In-Course Quick-Review Master Cheat Sheet

Feature / Parameter	Verbatim Fact on Page 41	In-Course Clinical Pearl
Experiment Number	Experiment 5	Effect of Route on Drug Effect
Full Title	Effect of Route of Administration on Drug Effect	Evaluates route-dependent pharmacokinetics
Aim	Find out effect produced when same drug given via different routes	Same drug (constant 100 mg/kg dose), varying routes
Animals	Rats -Male	Males eliminate estrous cycle hormonal bias
Apparatus	Syringes & needles, Oral feeding tube, stop watch	Gavage tube for oral, stopwatch for exact onset latency
Drug	Pentobarbitone or Nembutal	Short-to-intermediate acting oxybarbiturate
Molecular Target	\text{GABA}_\text{A}$ receptor complex	Increases chloride channel open duration
Initial Step	Weigh the first animal	Customizes volume to exact body mass
Standard Dose	100mg/kg	Hypnotic/anesthetic dose in rodents
Endpoint Test	Turn animal to lie on its back	Evaluates vestibular-spinal postural reflex
Endpoint Recorded	Time of loss of righting reflex (LORR)	Preclinical surrogate for general anesthesia
4 Routes Tested	a. Oral (100mg/kg), b. i/p, c. i/m, d. s/c	Enteral (1) vs Parenteral (3) comparison
Printed Q1	Time of onset of effect by various routes	Expected: IP > IM > SC > Oral
Printed Q2	Duration of the drug action	Time from LORR to recovery of righting
Printed Q3	Factors responsible for onset differences	Vascularity, surface area, first-pass metabolism